

BioPI: An Energy Efficient and Low-Latency Neuromorphic Pipelined System With Joint Design Optimizations of Sensor-Algorithm-Processor for Wearable Healthcare

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Abstract—Extensive research has been conducted on accurate classification of bio-signals for wearable healthcare applications, such as the detection of arrhythmia. Artificial Intelligence (AI) based methods utilizing dedicated neural networks (NN) have shown potential in classifying multi-class bio-signals. However, the separate designs of processors and sensors has introduced interaction overheads in existing studies, limiting their practicality and efficiency. Furthermore, there is a lack of detailed discussion on the joint design and optimization of bio-signal sensing, NN training, and NN acceleration. To address these challenges in wearable healthcare, this work proposes an efficient and low-latency neuromorphic system, BioPI, from a joint design perspective of sensor, algorithm, and processor. Inspired by biological neural systems, BioPI employs fully-spike dataflow with a hierarchical pipelined architecture, combining a mixed-signal analog-to-spike converter (ASC) and a compact pipelined spiking neural network (SNN) processor with event-triggered power gating. A unique duplicate-merge (DM) SNN algorithm is also presented to optimize the processing of ASC generated data during inference by converting the required SNN structures from 2-D to 1-D, which reduces on-chip memory size and operation numbers for inference by 2.02x and 1.98x, respectively, measured

on ECG classification task while maintaining state-of-the-art accuracy. To validate the system functionality, we built an end-to-end verification platform including a fabricated 40nm ASC chip and a FPAG implementation of the SNN processor. To evaluate the overall energy efficiency of the entire system on silicon, a complete design of BioPI is also implemented in 40nm process and post-layout simulations are conducted. In arrhythmia detection tasks using the MIT-BIH ECG dataset, BioPI achieves an overall accuracy of 97.5% in 4-class classification and an energy efficiency of 0.767uJ/classification, with a merely 12.54us execution latency.

Index Terms—Wearable healthcare, bio-signal processing, neuromorphic computing, spiking neural network, accelerator, system-on-chip, joint design of sensor-algorithm-processor.

I. INTRODUCTION

BIOMEDICAL signal processing techniques have proven to be promising in wearable health monitoring and disease detection applications [1], [2]. Research has demonstrated the effectiveness and ease of recording multiple bio-signals using wearable sensors [3], [4], [5]. Recently, artificial intelligence (AI) based systems have emerged for precise and scalable bio-signal classification tasks [6]. Dedicated deep neural networks (DNN), such as convolutional neural network (CNN) and recurrent neural network (RNN), have been proposed and implemented in edge healthcare monitoring systems [7], [8], [9]. To realize practical healthcare systems at edge, recent studies have shown a trend towards integrating both biomedical sensors and DNN processors within a compact system for real-time monitoring [10], [11], [12], [13].

The typical architecture of existing healthcare systems is illustrated in Fig. 1 (top). Captured biomedical signals are initially input into an analog-to-digital converter (ADC), where always-on sampling is utilized at a fixed frequency. The sampled multi-bit data sequences are then stored and transferred to the back-end digital signal processor (DSP) or neural processing unit (NPU), where DNNs are deployed for signal

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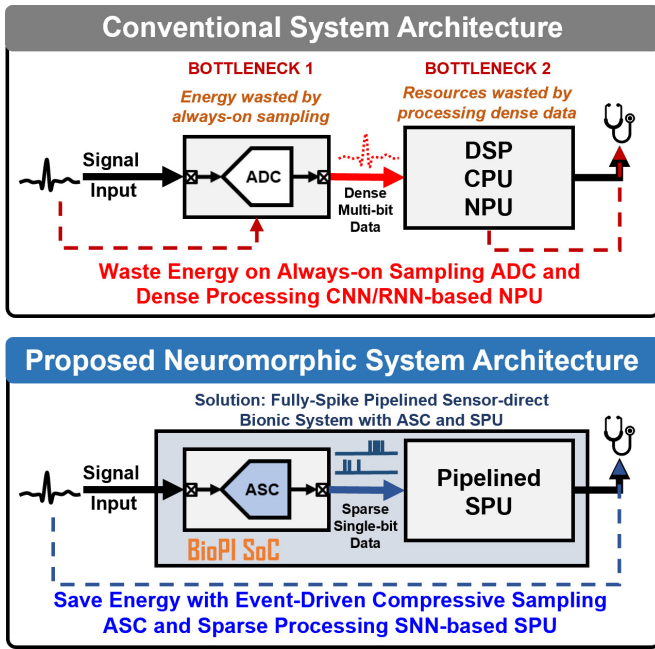


Fig. 1. Conventional system architecture and challenges (top); proposed neuromorphic system architecture and priorities (bottom).

classification and alert generation. Various optimization techniques have been proposed for both ADC design [14], [15] and biomedical NPU [16], [17] to enhance robustness and efficiency. However, such architectures suffer from two bottlenecks: (1) The gap between the separately designed sensor and processor leads to interaction overheads. The separation of the sensor and processor sets limits to the overall energy efficiency and response latency. (2) The power and memory overheads caused by sampling and processing dense multi-bit data in the sensor and processor, respectively. Conventional always-on sampling techniques convert analog signal to digital in a fixed frequency without considering the sparse nature of the biomedical signals, thus wasting energy for recording useless information. Additionally, data pre-processing modules and numerous multipliers are often required in the processor, which limits system energy efficiency.

Spike-based neuromorphic computing, inspired by the human brain's mechanism, has demonstrated significant potential in achieving energy efficient processing [18]. By emulating the neural dynamics in the brain, circuits and systems with high capabilities and energy efficiency become feasible. Various spiking neural networks (SNN) have shown potential in handling diverse healthcare tasks [19], [20], [21], [22], [23]. The highly sparse single-bit data representation enables SNN to execute with extreme resource limitations of power and memory on edge devices for different biomedical scenarios. The spike encoder plays a crucial role in SNN design, significantly influencing classification performance. However, most SNN-based systems are still unable to process data directly generated from biomedical sensors, incurring extra hardware costs in the implementation of spike encoder [24].

In our previous work [25], we proposed a fully-spike bio-signal processing system architecture that combines a level-crossing ADC (LC-ADC) and an SNN to demonstrate its potential. Several relevant works [26], [27] have focused on SNN processor designs with a digital LC encoder connected after a conventional ADC. However, existing reports overlook the interaction between the sensor-side data converter and the processor, and lack a detailed discussion about the co-design and co-optimization of the sensor, algorithm, and processor. Furthermore, reported SNN processor designs [26], [28], [29], [30] are based on reconfigurable processing elements. Although good flexibility is exhibited, such parallel-in-parallel-out architecture further increases inference latency, which has been a major drawback of SNN-based systems [31]. The remapping of different network layers also leads to more on-chip expense and lowers resource utilization ratio.

To overcome the challenges outlined above and enable ultra-efficient, application-specific edge healthcare systems, this paper proposes an energy efficient and low-latency neuromorphic system, BioPI, as shown in Fig. 1 (bottom). The whole system is designed with joint optimizations of sensor, algorithm, and processor. The BioPI system employs a bio-inspired fully-spike dataflow. The unique sensor interface acts as a compact integration of conventional ADC and digital spike encoder, converting continuous bio-signals into spike sequences, that are directly processed by the SNN model. A dedicated hardware-aware SNN training algorithm is designed to further compress the SNN model by merging 2-D SNN structures into 1-D. The SNN model is implemented on a pipelined flatten streaming SNN processor architecture that has a direct connection with the sensor. The entire system is implemented using 40nm CMOS process and verified on arrhythmia detection tasks using the MIT-BIH ECG dataset. The main contributions of this paper are as follows:

- We propose an end-to-end neuromorphic system from a co-design perspective of sensor, algorithm, and processor. The whole system combines a mixed-signal analog-to-spike converter and a layer-cascaded SNN processor with fully-spike dataflow to enhance system-level energy efficiency.
- We propose a duplicate-merge (DM) SNN algorithm, which is co-designed with the sensor-side sampling method. The proposed method achieves state-of-the-art accuracy while reducing on-chip memory size and operation numbers by 2.02x and 1.98x, respectively.
- We propose a hierarchical pipelined system-on-chip implementation to maximize on-chip resource utilization and decrease processing latency, which is parameterized and scalable.

The remaining sections of this paper are organized as follows. We detail the architecture and theory of BioPI in Section II. Section III illustrates the detailed hardware design and implementation. The effectiveness of BioPI system is evaluated and validated in Section IV. Lastly, the paper is concluded in Section VI where future potential is also discussed.

Joint Design Optimization Flow of Sensor-Algorithm-Processor

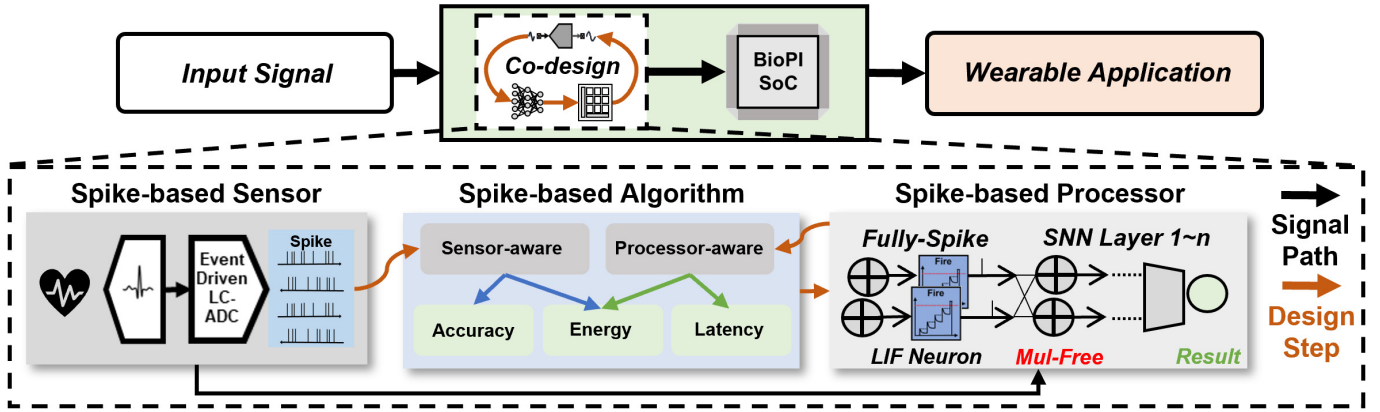


Fig. 2. Detailed dataflow of the BioPI system and the proposed end-to-end system design flow with joint design optimizations of sensor-algorithm-processor. Biomedical signals are collected from human body and sent into an analog-to-spike converter (ASC). The converted spike sequences are fed into the SNN processing unit (SPU) for classification. The SNN model is optimized with hardware-aware considerations.

II. ARCHITECTURE AND THEORY

Fig. 2 depicts the detailed dataflow of the BioPI system and the proposed end-to-end sensor-algorithm-processor co-design flow. For the system dataflow, raw biomedical signals are captured from the human body and sent into an analog-to-spike converter (ASC) based on a level-crossing ADC (LC-ADC), which acts as the spike encoder. The converted spike sequences are fed into the SNN processing unit (SPU) while the continuous raw bio-signals are processed by the ASC. All SNN layers are cascaded in the SPU and operate in a fully pipelined fashion. Since activations between layers of the SNN are binary data, no multiplication circuits are required in the SPU. After processing a complete period of input signal (e.g., a complete heartbeat), the classification result is sent to the user from the SPU. An alert will be raised if any abnormality is detected.

For the proposed joint design optimization of sensor-algorithm-processor, the design step starts from the spike-based sensor. The design specifications (e.g. resolution, signal margins, etc.) of the sensor circuits are decided according to the application requirements. The next step is to decide the SNN structure, where hardware costs of memory and energy consumption are taken into considerations. The last step is to design an optimal spike-based processor architecture to map the algorithm, which aligns with the spike-based sensor. To achieve an optimal balance between accuracy, energy efficiency, and processing latency, it is crucial to design the SNN model with consideration for the underlying hardware. Preserving the signal feature patterns within the spike-based sensor is essential for achieving high accuracy. Simultaneously, the SNN model should be designed to be compact, enabling reductions in power consumption and latency. To achieve this target, we propose a duplicate-training merge-inference SNN algorithm. This algorithm further reduces the computational burden and the size of the inference model. Additionally, we introduce the implementation of a scalable layer-cascaded pipelined processor with an event-triggered power gating unit. This implementation effectively reduces system-level latency and energy consumption.

The following subsections present the theories of our proposed methods in detail, including the event-based sensing principle, SNN processing dataflow, and the unique hardware-aware SNN training algorithm.

A. In-Sensor Analog-to-Spike Conversion

Biomedical signals, such as ECG, contain data features embedded within the temporal variance, which reflect the biological status within the human body. During the data sampling period, the signal amplitude remains almost unchanged in the silent periods. The absolute signal amplitude values during these periods have little relevance on distinguishing different signals. However, conventional frame-based Nyquist sampling ADC constantly samples data points with a fixed frequency, without considering this characteristic, leading to high sensor-side energy overheads and producing redundant multi-bit outputs. In contrast, level-crossing (LC) ADC is an event-based spike sampling data converter. Being sensitive to the temporal variance of input signals, the LC-ADC outputs spikes when predefined levels are crossed and outputs zero during the silent periods. In this way, the LC-ADC-based analog-to-spike converter (ASC) could reduce the sensor-side data sampling energy while preserving the temporal signal features for the next-stage analysis.

Fig. 3 provides a detailed comparison between frame-based sampling and event-based sampling. Notably, the outputs of ASC are sparse single-bit spike sequences, UP spikes and DN spikes, representing the up-crossing and down-crossing events, respectively. Thus, ASC achieves a high data compression ratio compared to frame-based Nyquist sampling ADC, resulting in reduced energy consumption of the sensor while retaining the most essential features within the signal.

To implement LC sampling in circuits, existing works connect a digital data encoder after conventional ADC to generate the spike sequences [26], [27]. However, this approach eliminates the superiority of LC-ADC in terms of event-triggered sampling and introduces more hardware overheads to encode

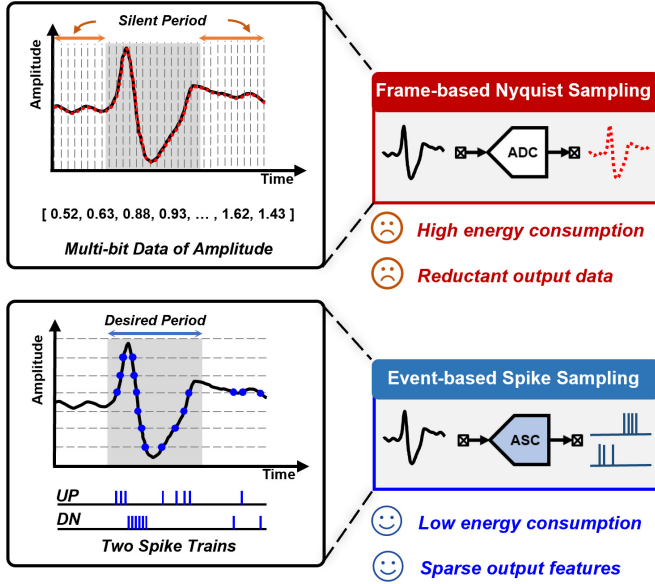


Fig. 3. Sensing principles of frame-based sampling with typical ADC and event-based sampling with LC-ADC.

the data before feeding it to the SNN. Other works apply spike event-based representation, which are known as the address-event-representation (AER), on auditory sensing [32] or vision sensing [33]. Although promising data compression is achieved, the output information still needs to be converted to frame-based sonogram images or common images for further processing, as the AER cannot be processed directly by the neural networks, leading to limitations on processing efficiency. Additionally, existing works must utilize UP spikes and DN spikes simultaneously for classification [26], [27], which can be computationally expensive. In this paper, we address these drawbacks by proposing a compact mixed-signal ASC circuit design for direct SNN processing. The proposed ASC converts raw analog signals into compressed digital spike sequences, which can be used as input activations of the SNN directly. In this work, we also design the SNN with a dedicated algorithm to merge the 2-D UP and DN spike vectors into 1-D spike vectors, achieving the same classification accuracy while decreasing the computational burden of classification by half. Further circuit details are provided in Part II-C.

B. SNN Processing Principle

The primary differences between deep neural network (DNN)-based artificial intelligence and spiking neural network (SNN)-based neuromorphic intelligence can be summarized in two key aspects: (1) SNN processes binary activations in the temporal domain, while DNN processes multi-bit activations in the spatial domain. (2) SNNs utilize bio-inspired neuron models like leaky-integrate-and-fire (LIF) neurons to emulate the behavior of biological dynamics, while DNNs typically rely on nonlinear activation functions such as rectified linear unit (ReLU) or sigmoid.

The detailed functionality of the LIF neuron is exhibited in Fig. 4. A LIF neuron receives a series of temporal spike inputs

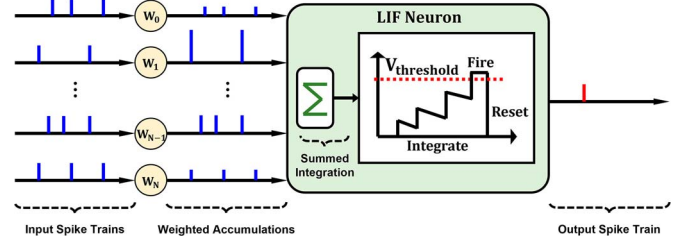


Fig. 4. Dataflow of spiking neural network (SNN) with a leaky integrate and fire (LIF) neuron.

and integrates these signals to update its membrane potential at each time step t . This membrane potential, denoted as $V[t]$, reflects the cumulative effect of incoming spikes. When $V[t]$ surpasses a defined threshold voltage V_{th} , the neuron generates an output spike and subsequently resets its potential to zero. In the presence of input spikes, the membrane potential accumulates; conversely, in their absence, it gradually diminishes due to a fixed leakage factor, termed $\mathcal{Leaky}$.

The mathematical representation of the membrane potential update for a LIF neuron is given by:

$$V_i[t] = \mathcal{Leaky} \times V_i[t-1] + \sum_j w_{ij} \times \delta_{ij}[t] \quad (1)$$

In this equation, $\delta_{ij}[t]$ represents the pre-synaptic spike inputs from the previous layer, while w_{ij} denotes the corresponding synaptic weights. The index i refers to the neuron of interest, and j indicates the neurons connected to the preceding layer. This framework allows SNNs to capture temporal dynamics, offering a more biologically plausible approach to processing information compared to traditional DNNs.

This event-driven mechanism enables SNNs to operate efficiently, particularly in scenarios where input data is temporal time-series, such as in sensory processing for medical wearables. Unlike DNNs, which require continuous activation functions and often suffer from high power consumption, SNNs can remain idle during periods of inactivity, thus conserving energy. The spike-based communication in SNNs also introduces a level of robustness to noise, as information is conveyed through discrete spike events rather than continuous values. This characteristic is particularly advantageous in medical applications, where signals can be affected by various sources of interference. Furthermore, the temporal coding in SNNs allows for rich representations of dynamic inputs, enabling the model to capture time-dependent patterns that are crucial for tasks such as patient monitoring and anomaly detection.

C. Duplicate-Training Merge-Inference Algorithm

The spike sequences processed by SNN are directly generated from the proposed ASC. The SNN is first trained and then deployed for inference. It is noteworthy that there are two kinds of features generated from ASC: the UP spikes and DN spikes representing the up-crossing and down-crossing of input signals, respectively. Popular methods directly utilize these two spike sequences together with direction information embedded,

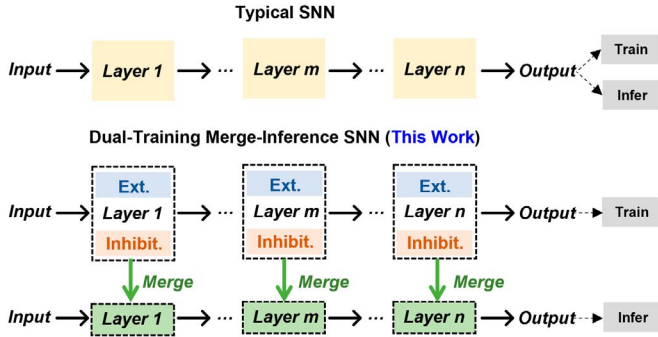


Fig. 5. Network structures of typical SNN and dual-training merge-inference SNN in this work.

which requires a neural network with 2-D kernels for processing [26], [27], [28]. However, edge devices are quite sensitive to on-chip resources, including memory size and computation complexity. Compared with 1-D neural networks, 2-D neural networks requires more weight memory and more computation resources. The required operation numbers in 2-D network is also more than 1-D. Thus, 1-D neural network with 1-D kernels and 1-D inputs has demonstrated superiority in efficiency in terms of memory and computation against 2-D networks. Merging the UP and DN spike sequences from ASC could transfer the inputs from 2-D to 1-D but eliminates the directional information, which degrades the classification accuracy. Motivated by recent findings in neuroscience revealing the presence of excitatory neurons and inhibitory neurons within the neural circuitry [34], dual-neuron SNN has been studied to enhance the classification accuracy in various tasks. This enhancement is obvious since dual-neuron SNN with inhibitory neurons or inhibitory layers avoids the information loss caused by adopting positive-only activations in typical SNN [35]. This finding motivates us to design a SNN to perform accurate classification with the merged spike sequences. However, dual neuron structures double the required operation number at the same time. To ensure both computation efficiency and accuracy, we propose the duplicate-training and merge-inference (DM) algorithm to process spike data from ASC with SNN. The comparison of our DM-SNN with existing SNNs is shown in Fig. 5. In typical single-neuron SNN and dual-neuron SNN, the training and inference are based on the same SNN structures [18], [35]. In our proposed DM-SNN, the training and inference are separated to realize both accurate training and efficient inference. In the training phase, a dual-dendrite single-soma SNN structure is adopted to distinguish the UP and DN spikes within a merged 1-D input sequence. Conceptually, excitatory dendrites apply positive accumulations, and inhibitory dendrites apply negative accumulations to the membrane potential of soma, which is relevant to the up-crossing events and down-crossing events of input signals in ASC. This is possible as the output of our SNN is decided by the firing rates of neurons, thus the expected neural connections are formulated during training. Although the UP and DN spikes are merged within the inputs, the weights of excitatory dendrites and inhibitory dendrites are trained to extract the directional features via spatial and

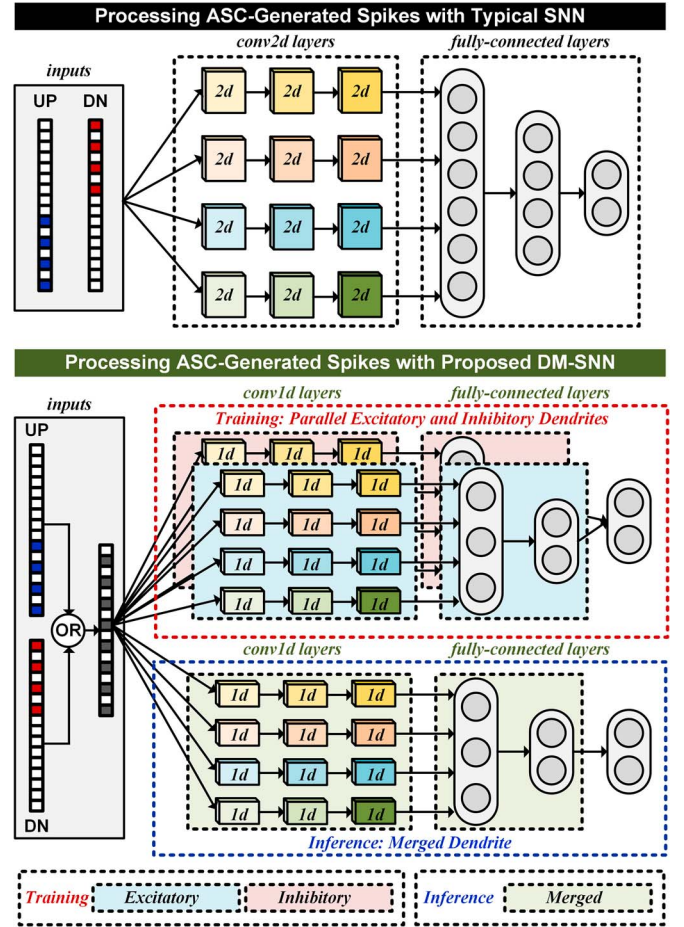


Fig. 6. Comparison of processing ASC-generated spikes with typical SNN and our proposed DM-SNN.

temporal back-propagation. In the inference phase, the dual-dendrite structure is merged into a single-dendrite fashion to reduce the computation cost.

A detailed illustration of comparison between typical SNN and the proposed DM-SNN is shown in Fig. 6. When processing ASC-generated spikes with a typical SNN, 2-D spike inputs containing both UP and DN are fed into 2-D convolution layers first (*conv2d*) and then fully-connected layers. The same SNN structure is utilized for both training and inference in a typical SNN. In DM-SNN, the spike inputs are 1-D sequences by merging UP and DN with a OR gate. The merged sequences are fed into 1-D convolution layers (*conv1d*) instead of *conv2d* in a typical SNN. As activation spikes across layers are reduced from 2-D vectors to 1-D vectors, the neural connections within fully-connected layers are also reduced by half. As mentioned before, a parallel excitatory and inhibitory neural structure is applied in the training phase of DM-SNN. After training, the parallel neurons are merged together for low-cost inference. Comparing the inference structures of typical SNN and our method, DM-SNN brings two advantages when processing ASC-generated spike data: (1) by merging the input UP and DN spikes and applying 1-D network structure during the inference, the required on-chip memory size and total operation numbers

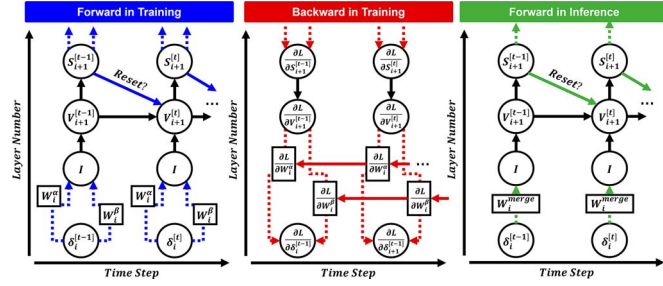


Fig. 7. Spatial-temporal dataflow of training and inference in the proposed DM-SNN.

are reduced by approximately 50%, respectively; (2) validated on ECG-based arrhythmia detection tasks, less than 1% accuracy loss incurs in inference with 1-D SNN by adopting DM-SNN training method compared to inference with 2-D SNN.

Fig. 7 illustrates the detailed dataflow of both training and inference in DM-SNN. The training phase consists of a forward path (FT) and a backward path (BT). The BT is organized in a backpropagation-through-time (BPTT) fashion. The inference phase contains forward path only (FI). As mentioned before, both FT and BT are conducted based on parallel excitatory and inhibitory dendrites, while FI is conducted based on merged dendrites.

When performing FT, inside every layer of the proposed SNN, each LIF neuron consists of an excitatory dendrite α and an inhibitory dendrite β that parallelly receive the spike activations from the previous layer. Within the forward path, when the membrane potential is updated in the neurons, the accumulation of pre-synaptic stimulus I for the corresponding LIF neuron is expressed as:

$$I = \sum_j w_{ij}^\alpha \times \delta_{ij}[t] - \sum_j \mu \times w_{ij}^\beta \times \delta_{ij}[t], \quad (2)$$

where w_{ij}^α denotes the weight on the excitatory dendrite, and w_{ij}^β denotes the weight on the inhibitory dendrite. μ is a hyper parameter controlling the inhibition density, which is fine-tuned to optimize the classification performance. Thus, the updated mathematical model of LIF membrane potential can be expressed as:

$$V_i[t] = \text{Leaky} \times V_i[t-1] + I, \quad (3)$$

And the output spikes transferred to the next layer can be generated:

$$S_{i+1}[t] = (V_i[t] > V_{th}) ? 1 : 0 \quad (4)$$

After the input spikes go through all the layers within the SNN, the final output spikes o can be obtained. In DM-SNN, the error function of BT is mean square error (MSE). To calculate the MSE loss, the output spikes of the final layer o are accumulated throughout all the time steps (T) to measure the firing rate. The neuron index with the maximum firing rate indicates the classification result. Meanwhile, the sample labels are provided in one-hot format. With the expectation to only fire the neuron

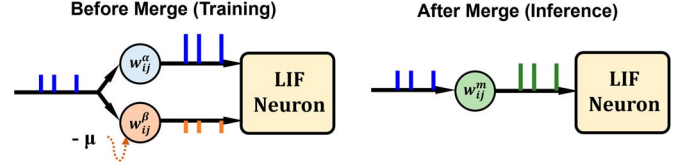


Fig. 8. Neural dataflow before and after merging in the proposed DM-SNN.

indexed as the corresponding label, the loss function L can be expressed as:

$$L = \frac{1}{2N} \sum_{n=1}^N \left\| y_n - \frac{1}{T} \sum_{t=1}^T o^{n,t} \right\|^2, \quad (5)$$

where N is the total number of training samples and T is the total time steps. y_n represents the corresponding sample label in one-hot fashion.

When performing BT, the weights of each layer are updated through BPTT of the loss (Fig. 7 middle). Notably, the weight updates of SNN are calculated by applying the chain-rule both spatially and temporally. Spatially, the derivatives of the loss to the weights of LIF dendrites are calculated layer-by-layer. Temporally, the final derivatives are jointly decided by the neural dynamics throughout all the time steps by accumulating the time-step-wise update amounts. Applying chain-rule throughout all the layers and time steps, the mathematical expression of weight update is provided as:

$$\delta w_i = - \sum_{t=1}^T \left(w_{i+1}^T \times \frac{\partial L}{\partial V_i[t]} \right) \times f'(V_i[t]) \times V_i[t]^T \quad (6)$$

After the training is finished in BT, all the weights are fixed for further operation in FI. As mentioned before, FI is conducted based on merged excitatory and inhibitory dendrites. This is possible as the weights on the excitatory dendrite (w_{ij}^α) and the inhibitory dendrite (w_{ij}^β) share the same vector size and dimension. Furthermore, as each pair of the excitatory dendrite and the inhibitory dendrite receives the same input spikes and contributes to the same accumulation of membrane potential, this behavior can be replaced with a merged dendrite within the pair. Thus, every dendrite pairs connecting to the same LIF neuron can be merged together in the inference phase as follows:

$$w_{ij}^{merge} = w_{ij}^\alpha - \mu \times w_{ij}^\beta \quad (7)$$

This can help to further achieve efficient inference and reduce the resource in hardware implementation. Fig. 8 details the comparison of neural dataflow before and after merging.

III. SYSTEM IMPLEMENTATION

To support efficient implementation of the BioPI system-on-chip (SoC), we propose an end-to-end architecture that combines a mixed-signal analog-to-spike converter and a layer-cascaded SNN processor with fully-spike dataflow. The configurations of our SNN processor design is decided by the utilized network for ECG processing, while it's also scalable to various

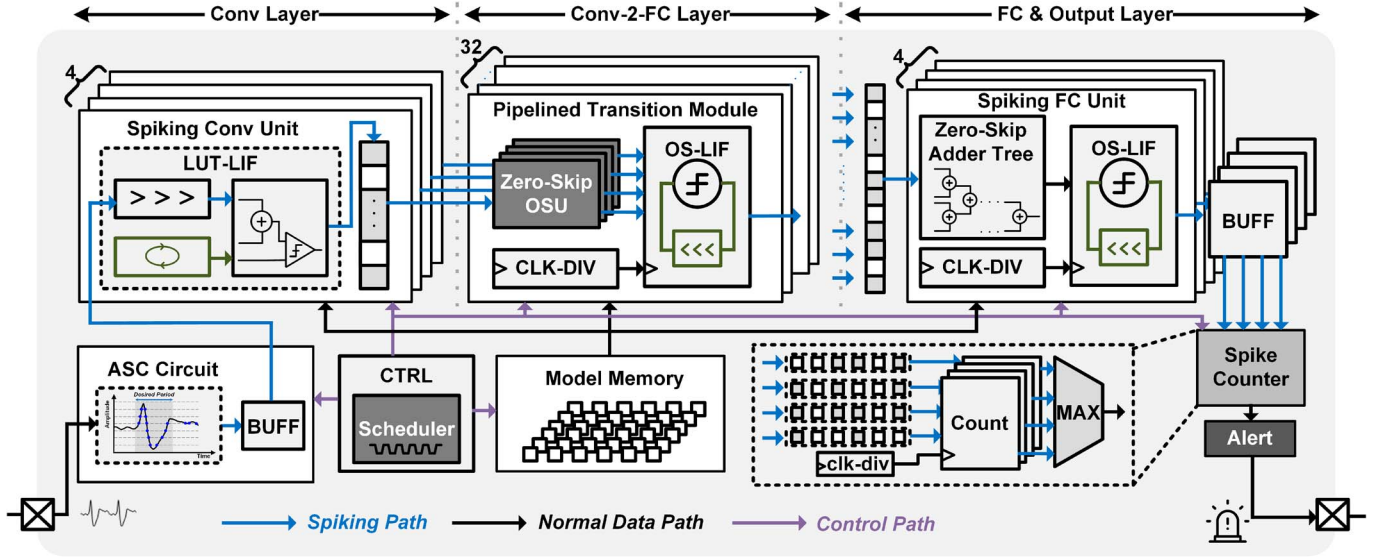


Fig. 9. Overall hardware architecture of BioPI SoC. The spiking path illustrates the dataflow for a complete inference, while the normal data path depicts the loading of model parameters. The control path indicates the source of control signals.

SNN structures for other applications. We also propose a layer-cascaded pipeline to maximize on-chip resource utilization and decrease processing latency, which is a significant drawback of SNN-based systems to process input activations across multiple time steps. The entire BioPI SoC is implemented in a 40nm CMOS process and verified with a mixed ASCII/FPGA platform on arrhythmia detection tasks using the MIT-BIH ECG dataset. The following subsections provide details of the proposed hardware designs.

A. Overall Hardware Architecture of BioPI SoC

Fig. 9 depicts the overall hardware architecture of BioPI SoC. The input raw bio-signal is first fed into the ASC circuit, which generates the 1-D spike sequences. The spike data from ASC is then sent to the SPU, where a cascaded 3-layer SNN is implemented for ECG processing in our validation. The SPU comprises three modules: spiking convolution unit (SCU), pipelined transition module (PTM), and spiking fully connection unit (SFC). The output generation module is composed of spike counters connected to the outputs of SFC. The spike counter is built using counters and comparators to accumulate spikes from the output neurons and identify the maximum count. This generates the output classification results by indicating the index of the output neuron with the highest firing rate. The entire model weights are stored on-chip and sent to SCU, PTM, and SFC, respectively, during inference. A global controller (CTRL) controls the timing of all the modules. The 3-layer SNN operates in a layer-cascaded pipelined fashion. The detailed hardware implementations and SNN mapping are illustrated in the following sections.

B. Analog-to-Spike Converter

Fig. 10 illustrates the circuit implementation of the ASC with a synchronous interface. The ASC is implemented based

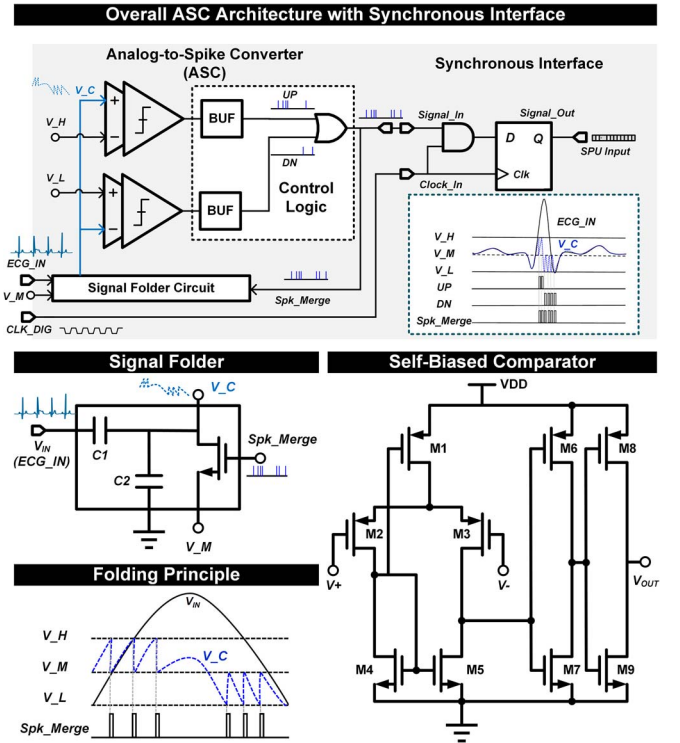


Fig. 10. Mixed-signal analog-to-spike converter circuit with a synchronous interface. The merged output spikes are sent to the following SNN processor.

on the LC-ADC proposed in [15]. The ASC consists of three key modules: a signal folder circuit, low-power self-biased comparators, and control logic circuits. Fig. 10 (top) illustrates an example waveform illustration of the input ECG signal. As shown in Fig. 10 (bottom), the signal folder circuit, based on a 1-bit capacitive DAC, processes the input analog signal along with the asynchronous trigger (Spk Merge) folding the input

to a fixed voltage (V_M) through charge redistribution. This design ensures charge injection remains independent of the input signal, with offsets eliminated via calibration. The comparator shown in Fig. 10 (bottom) uses a 3-stage design with a self-biased differential amplifier, followed by inverter stages for rail-to-rail digital output. A negative feedback loop stabilizes bias current, reducing power consumption by eliminating the need for an external bias circuit. Most transistors operate in the sub-threshold region to save energy. Additionally, the control logic processes comparator output spikes to generate asynchronous trigger feedback (Spk Merge), using inverter chains to enhance slew rate and noise tolerance. Standard technology library cells are used to implement the control logic circuits. The synchronous interface circuits are designed for both testing and interaction with the proposed SPU processor. An external digital clock is used to synchronize the output spikes, aligning the spike pattern with the timing of digital processing. The UP and DN sequences are merged and fed into the SPU processor.

The pulse length of ASC output spikes is determined by the combined delays of the signal folding circuit, comparators, and control logic circuits, as derived from the working mechanism of the ASC. Specifically, when the input signal exceeds the detection window defined by V_H and V_L , it triggers an output spike (UP/DN) to go high. A feedback signal is generated after a delay introduced by a logic OR gate, which activates the switch in the signal folding circuit for charge redistribution. Once the input signal is folded back to the detection window, the output of the comparators goes from high to low, causing the ASC output spike to go low. Therefore, the pulse length of ASC output spikes is the sum of the charge redistribution delay of the signal folding circuit, the delay associated with signal comparison related to the comparator's slew rate, and the delay of logic gates and buffers in the comparator and control logic circuits.

In this work, the spike pulse length of the proposed ASC is designed to accommodate the varying sampling rate requirements of different points in the input signal, especially the maximum input variance that produces the minimum time between output pulses. For illustration, consider an input sine wave $V(t)$ with frequency f_{in} and amplitude A_{in} . The lowest signal change rate occurs at the peak and valley points, while the highest signal change rate is observed at the zero-crossing points. By taking the derivative of the input signal, one can determine the input change rate. If the peak-to-peak amplitude of the input signal matches the full-scale amplitude of the ASC, the value of the LSB for an M-bit ASC is given by:

$$LSB = \frac{2A_{IN}}{2^M} \quad (8)$$

Thus, the minimum time between the sampled data T_{min} is equal to:

$$T_{min} = \frac{V_{LSB}}{\left. \frac{dV(t)}{dt} \right|_{\cos(2\pi f_{IN}t)=1}} = \frac{1}{2^M \pi f_{IN}} \quad (9)$$

The parameter T_{min} is a crucial in circuit design, as it determines the maximum spike pulse width and ensures a rapid response to the maximum input change rate. For instance, a

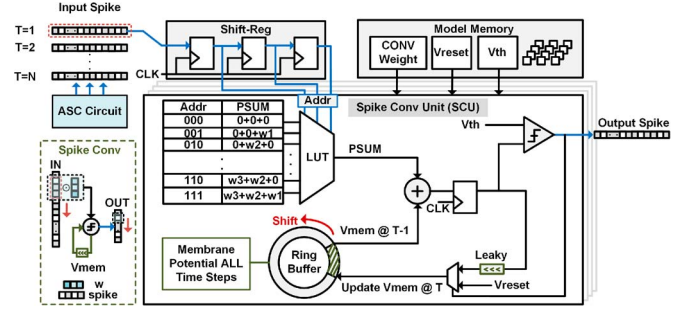


Fig. 11. The computing flow of spiking convolution operations (Spike Conv) and detailed circuits of spike convolution units (SCU).

6-bit ASC with a maximum input frequency of 10 kHz would require a T_{min} value of 497.4 ns. As such, the spike pulse width and propagation delay of circuit modules must be less than this value to meet the required speed specifications.

The final bit length of the digital spikes for SPU inputs is jointly controlled by the ASC spike pulse length and the external digital clock. The external digital clock CLK_{DIG} drives the interface buffers, which sample ASC spike pulses into digital spike bits. The frequency of CLK_{DIG} is significantly higher than the input signal, ensuring the stability of the SPU inputs. However, the direct output sequences of the interface buffers contain redundant 1/0 bits as the digital interface samples multiple times during one ASC generated pulse, which are not desired. To resolve this issue, the output sequences are further compressed with fixed-port OR gates to achieve a pre-defined fixed length.

C. Spiking Convolution Unit

The spiking convolution unit (SCU) circuit design and data mapping for spiking convolution operations are illustrated in Fig. 11. This work employs 1×3 convolution kernels, with four channels of convolution kernels operating in parallel for the case of ECG processing. At each time step, a 256-bit spike sequence is serially fed into the SCU. The spike sequence is then swept using the four convolution kernels, and the partial sum (PSUM) of each convolution operation is integrated in the corresponding leaky integrate-and-fire (LIF) neuron to generate the output spikes. These output spikes are then forwarded to the next-layer operations.

To optimize the SCU implementation at the circuit level, several techniques are employed. The input spikes are received by a 3-bit shift register. As the kernel size is relatively small, all the possible PSUM values are pre-stored in a look-up table (LUT), where the 3-bit output from the shift register serves as the address (Addr). By adopting the LUT-based PSUM generation scheme, both processing latency and hardware cost are reduced compared to those using arithmetic operators like parallel adders. This optimization is achieved since very tiny convolution kernels are adopted in this work. If the network scales up with bigger kernel size, the LUT modules in SCU can be replaced by adder trees to realize the similar pipeline.

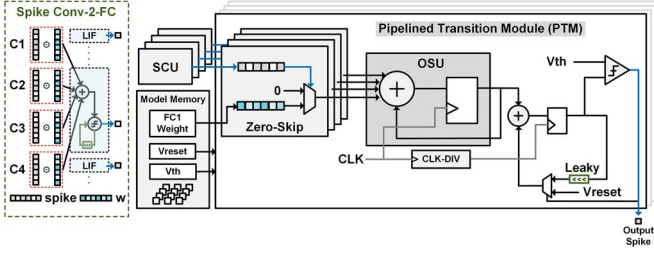


Fig. 12. The computing flow of the flattened layer followed by the first spiking fully-connected layer (Spike Conv-2-FC) and detailed circuits of the pipelined transition module (PTM).

Within the convolution layer, each convolution kernel is associated with a LIF neuron circuit. Apart from PSUM, the LIF neuron circuit also receives the membrane potential. In existing designs [26], [28], the membrane potentials are fetched from separate memories and buffered before feeding into the computation cores, which increases both latency and hardware overheads. In our proposed SCU, the LIF neuron circuit receives the PSUM from the LUT and the membrane potential from a ring buffer. The ring buffer has a read index and a write index, which pipelines the fetching and updating of membrane potentials. This design eliminates the overheads of extra buffers between separate membrane potential memories and computation cores in existing reconfigurable processing element based designs [26], [28]. Driven by the digital clock (CLK), the membrane potential V_{mem} is updated and compared with the threshold potential V_{th} . If a spike is fired at the output, the membrane potential V_{mem} is reset to V_{reset} ; otherwise, it leaks via right shifting and writes back to the ring buffer. Notably, the leakage factors in our SNN are all factors of 2. The read index and write index of the ring buffer also shift as the input spikes come into the SCU to sweep through all the membrane potentials in the convolution layer. After receiving the complete input spike sequence, SCU finishes all the spike convolution operations in this time step, and the read index of the ring buffer goes back to the initial position.

D. Pipelined Transition Module

The pipelined transition module (PTM) serves as the first fully-connected layer in the SNN, which receives the output spikes from the SCU. Fig. 12 illustrates the detailed dataflow with circuit structures. In SNN, the multi-channel data needs to be flattened before the first fully-connected layer can receive outputs from the last convolution layer. In conventional homogenous or flattened streaming NN cores, the outputs of the last convolution layer are sent to the activation memory and transferred to the flattened layer circuits, resulting in higher latency and more hardware costs. In BioPI, we propose PTM circuits to embed the flattening operation within the first fully-connected layer.

The first fully-connected layer in SNN accumulates all the received spikes in each LIF neuron in parallel. Therefore, PTM receives outputs from four SCUs simultaneously. The corresponding weights are accumulated in a zero-skipping fashion

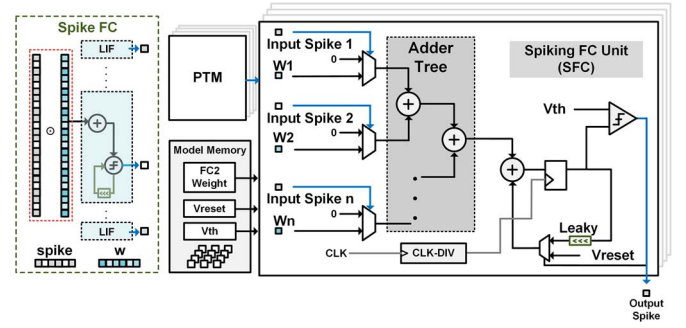


Fig. 13. The computing flow of the following spiking fully-connected layer (Spike FC) and detailed circuits of the spike fully-connected units (SFC).

[36]: if the spike is 1, the weight is sent in; otherwise, 0 is sent out. The four zero-skip units receive four spikes from the four SCUs at each clock cycle, and the accumulations are summed up in the output-stationary unit (OSU). A counter is shared by all the PTMs to record the number of received spikes. Once the OSU receives all the output spikes from the SCU in one time step, the stationary output in OSU is reset to zero. A divided clock signal CLK_{DIV} is also generated via the same counter to drive the LIF neuron circuit connected with OSU, which ensures that the membrane potential is correctly updated after receiving all the spikes from the SCU in every time step. The LIF neuron circuit then generates the output spike and sends the spike to the next fully-connected layer. In this work, the first fully-connected layer has 32 neurons, and thus there are 32 PTMs working in parallel. Notably, it is convenient to scale up the number of neurons by duplicating additional parallel PTM circuits connected to SCUs. The clock frequency can remain unchanged as no overhead is introduced in the critical path of computation, but the total area and power consumption of PTMs will increase linearly.

E. Spiking Fully-Connection Unit

The other fully-connected layers are implemented in the SFC circuits, as illustrated in Fig. 13. The second fully-connected layer receives the output spikes from PTMs. The corresponding accumulations are generated simultaneously using the adder trees in the same zero-skipping fashion as PTMs. The accumulation results are then sent to the corresponding LIF neuron circuits. The LIF neuron circuits in SFC are driven by the same CLK_{DIV} signal as PTMs. In this work, the second fully-connected layer is the last layer of our SNN, and there are 4 neurons mapped on 4 SCUs in total. Each SCU is connected to all the 32 PTMs and sends out the spikes to the spike counter for obtaining the final classification results.

F. Event-Triggered Power Gating Unit

As biomedical signals always have relatively low frequency, most of the processing time is consumed waiting for the signal recording, which refers to the stand-by mode in our design. If the SPU always keeps active, the leakage power of the SPU will dominate the SoC energy consumption during signal recording

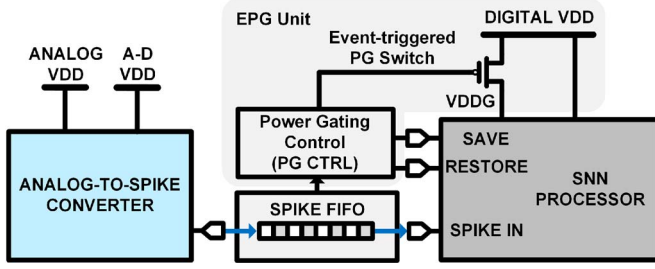


Fig. 14. The event-triggered power gating (EPG) unit and control paths.

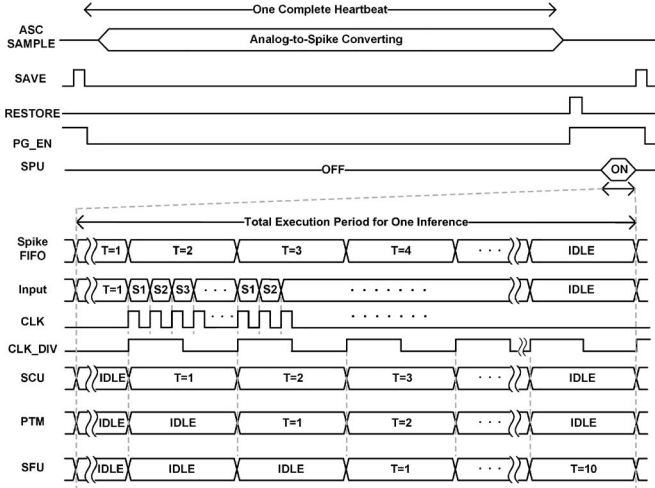


Fig. 15. Timing analysis of the proposed pipeline flow between different modules in BioPI.

periods, which is not expected. To reduce power consumption in stand-by mode, we propose an event-triggered power gating (EPG) unit to adaptively power on and power off the SNN processor. As shown in Fig. 14, the power gating schedule is controlled according to the status of the spike first-in-first-out (FIFO) unit and the outputs of SNN processor. The spike FIFO sends out a flag to the PG CTRL after receiving all the required spikes. Upon receiving the flag, PG CTRL commands the SNN processor to enable the power gating cells. After completing the entire classification, the SNN processor notifies the PG CTRL to disable the power gating cells. To retain the SNN weights and parameters during power off, dual control balloon retention (DCBR) registers are utilized to store that data instead of common registers. The DCBR register contains one more latch compared to common data flip flop (DFF). The latch consists of minimum-sized transistors and operates in always-on mode. Configured by the control signals SAVE and RESTORE, the DCBR register saves the stored data into the latch before power off and reads the data back after power on. In this way, the total power during stand-by mode is greatly reduced.

G. Timing Analysis of BioPI

Fig. 15 presents a detailed timing analysis of our proposed event-triggered power gating and hierarchical pipeline schemes.

The BioPI SoC operates in the stand-by mode after initialization, retaining the programmed data, while the ASC generates spike sequences from the input signal. Once the complete heartbeat is recorded, the power gating cells are enabled, and the SNN processor enters the operation mode. In this mode, the entire BioPI system operates in a fully-pipelined fashion to maximize the on-chip resource utilization ratio and minimize processing latency issues in SNN processors. The consumed inference time is the delay between receiving the input spike and generating the output result. The number of time steps is set as a programmable number according to the SNN model.

In the first time step, the spike FIFO sends out spikes into the input buffers of the SNN processor, and other modules remain in an *IDLE* state. In the second time step, driven by the *CLK* signal, SCUs receive the spike sequence from input buffers and perform convolution computations while the buffers receive the next spike sequence. In the third time step, all the convolution outputs are sent to PTMs, and SCUs receive the spike sequence generated in time step 2. In the fourth time step, driven by the *CLK_DIV* signal, SFU generates the output spikes and sends them to the spike counter. This pipeline process continues until all the spikes inside the spike FIFO are processed. The spike counter then produces the classification result and raises an alert if an abnormality is detected. Using our proposed pipelined hardware architecture and timing design, all the SNN layers operate in a cascaded fashion, and the total required processing time is significantly reduced. It is noteworthy that our proposed pipelined hardware architecture and timing design can be easily scaled to various model structures, which also enables agile design for broader edge computing scenarios.

IV. EXPERIMENTS AND RESULTS

In this section, we showcase the capabilities of BioPI by prototyping an electrocardiograph (ECG) classification system to monitor cardiac arrhythmia, which is a common cause of death globally, as reported by the World Health Organization [37]. First, we introduce the experimental setup, encompassing both software and hardware, and subsequently provide a comprehensive analysis of our system's performance and the comparison with state-of-the-art works.

A. Experimental Setup

The evaluation of BioPI in ECG classification is based on the MIT-BIH dataset [38], a commonly used benchmark for arrhythmia detection tasks. To validate the system functionality, we built an end-to-end verification platform including a fabricated 40nm ASC chip and a FPAG implementation of the SNN processor. To evaluate the overall energy efficiency of the entire system on silicon, a complete design of BioPI is also implemented in 40nm process and post-layout simulations are conducted. The weights of the SNN are trained based on the spike sequences generated from the ASC using PyTorch.

B. ASC Under PVT Variations and Mismatch

To assess the performance and robustness of the ASC chip under PVT variations, we conducted 3-corner (TT, SS, FF)

TABLE I
ASC PERFORMANCE UNDER PVT VARIATIONS

Corner	TT	TT	TT	TT	SS	FF
Supply (V)	0.5	0.5	0.45	0.55	0.5	0.5
Temperature ($^{\circ}\text{C}$)	-20	80	27	27	27	27
Bandwidth (kHz)	10	10	10	10	10	10
SNDR (dB)	39.0	28.7	37.6	40.3	33.6	39.4
ENOB (bit)	6.2	4.5	6.0	6.4	5.3	6.3
Power (nW)	79.7	219.9	85.1	187.2	65.6	672.5

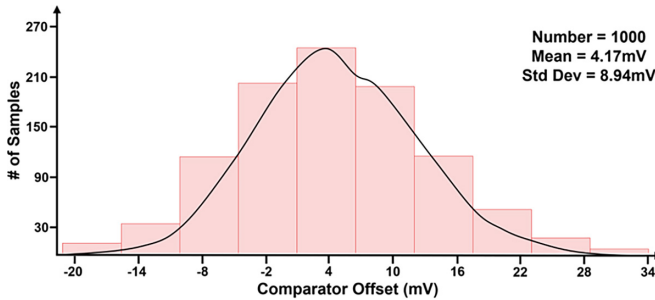


Fig. 16. 1000-runs Monte Carlo simulation on post-layout comparator input offset mismatch.

simulations, incorporating temperature and supply voltage variation during the post-layout simulation. The results of these corner simulations are summarized in Table I. As shown, the ASC exhibits robust performance with expectation to achieve 5 bits ENOB despite PVT variations. While the ASC achieves only 4.5 bits ENOB at the TT corner at 80°C , it is important to note that such high temperatures are not typically encountered in wearable healthcare scenarios. In fact, under actual measurement conditions in a laboratory environment at room temperature, the fabricated ASC achieved the ENOB of 6.7 bits with the same input frequency settings, demonstrating its reliable performance.

The comparators used in the ASC are designed with sub-threshold transistors, which may experience mismatch that could potentially affect the detection window defined by V_H and V_L . Fig. 16 shows 1000-runs Monte Carlo simulation on post-layout comparator input offset mismatch. The simulation results show that the mean value of input offset is 4.17mV with a standard deviation of 8.94mV. To address this concern, we utilized a triangle wave with an amplitude of 640mV and a frequency of 1kHz during the chip testing process to calibrate the effect of the mismatch and determine the right detection window thresholds to ensure that the fabricated ASC generates the expected outputs of UP and DN spike sequences.

C. Dataset Preparation

According to the standards set by the Association for the Advancement of Medical Instrumentation (AAMI) [39], ECG recordings can be categorized into five classes: ventricular ectopic beats (VEB), supraventricular ectopic beats (SVEB),

fusion beats (F), non-ectopic beats (N), and unknown beats (Q). The MIT-BIH dataset includes 90081 N samples, 7008 VEB samples, 2781 SVEB samples, 802 F samples, and 15 Q samples. Since the sample amounts vary significantly between different categories, the data imbalance issue will affect the integrity of reported classification performance. In our experiments, to address the issue of data imbalance and conduct a fair evaluation, we randomly select 8000 samples from N beats, and align the number of VEB, SVEB, and F beats to 8000, respectively, with random duplication. Q samples were excluded from this experiment the same as other approaches due to their limited number, which made them unsuitable for training. The total of 32000 samples was then randomly divided into training and testing datasets in a 4:1 ratio.

To effectively test the performance of the BioPI SoC using the MIT-BIH dataset, ECG samples, which are represented by digital values, are first reconstructed into continuous analog signals before being sent to the ASC. The original ECG recordings were digitalized at a sampling rate of 360Hz with 11-bit resolution and a range of 10 mV. To convert these digital recordings into continuous analog signals, we assume that the signal values varies linearly between each pair of digital data points. Based on this principle, all continues analog ECG samples are firstly produced by a signal generator and then fed into a 5-bit ASC to generate the corresponding spike sequences. These generated spike sequences form a complete spike dataset, which is then utilized for SNN training and testing. In our experiments, each spike sample contains 2500 spikes, divided into 10 time steps. This work represents one of the first efforts to generate spike-based ECG dataset from raw signals with a realistic ASC, while existing works [16], [23], [26], [27] typically employ digital encoding algorithms to generate spikes from digital samples within the original MIT-BIH dataset. [40] also demonstrates a good example of generating spike-based auditory data with a mixed-signal approach. Future works are expected to explore mixed-signal spike conversion designs for various signal processing scenarios.

D. Classification Performance

We developed two models, namely the 2-D SNN and 1-D SNN, to evaluate the performance of our proposed DM-SNN algorithm. Both SNNs consist of one convolutional layer, one flatten layer, and two fully-connected layers. The 2-D SNN receives 2-D spike sequences, including both UP and DN, while the 1-D SNN receives only the merged 1-D spike sequence from the ASC. Accordingly, the 2-D SNN utilizes 2×3 convolution kernels, while the 1-D SNN employs 1×3 convolution kernels, with the input channel set to 1 and the output channel set to 4 for both models. The flatten layer is designed based on the shapes of outputs from the convolutional layers of the 2-D SNN and the 1-D SNN, respectively. The two fully-connected layers in both models share the same structure, with the first layer consisting of 32 neurons and the second layer consisting of 4 neurons. The 2-D SNN is trained using the typical BPTT algorithm, while the 1-D SNN is trained using our proposed DM-BPTT algorithm. All other hyperparameters are fine-tuned

TABLE II
SUMMARY OF DETAILED ECG CLASSIFICATION PERFORMANCE AND
COMPARISON WITH STATE-OF-THE-ART

Method	ECG Type	Precision (%)	Recall (%)	Specificity (%)
TBioCAS 2020 [16]	N	-	-	-
	VEB	0.98	-	-
	SVEB	0.943	-	-
	F	-	-	-
TBioCAS 2023 [17]	N	-	-	-
	VEB	0.982	-	0.998
	SVEB	0.873	-	0.999
	F	-	-	-
Our 2-D SNN (Typical BPTT)	N	0.991	0.991	0.997
	VEB	0.981	0.969	0.994
	SVEB	0.966	0.984	0.989
	F	0.993	0.987	0.998
Our 1-D SNN (DM-BPTT)	N	0.994	0.975	0.992
	VEB	0.959	0.95	0.986
	SVEB	0.968	0.981	0.99
	F	0.997	0.991	0.999

to obtain optimal results. Table II summarizes the classification performance of our 2-D SNN and 1-D SNN, respectively, along with a detailed comparison with state-of-the-art works. As demonstrated, both models exhibit competitive performance in terms of precision, recall, and specificity. Notably, the 1-D SNN achieves nearly lossless results compared to the 2-D SNN, thereby validating the effectiveness of our proposed DM-SNN algorithm.

E. Improvement Analysis

To quantify the improvements achieved by our proposed method, we recorded the on-chip memory (MEM) occupancy and the number of operations consumed (OPS) for both the 2-D SNN and 1-D SNN models described above. The recorded MEM includes both the activation buffer size and the complete SNN parameters, formulated as below:

$$MEM_{SNN} = \sum_{layers} (Weights \times WeightPrecision + Neurons \times V_{MEM}Precision + Activations \times T) \quad (10)$$

The measured OPS numbers are calculated as follows:

$$OPS_{SNN} = \sum_{CONV} M_H \times M_W \times (K_H + K_W - 1) \times C_{in} \times C_{out} \times T + \sum_{FC} N_{in} \times N_{out} \times T \quad (11)$$

where M , N , K , C , H , W , and T represent the feature map in the convolution layers (CONV), connection numbers in the fully-connected layers (FC), convolution channels, the convolution kernels, kernel height, kernel width, and time step numbers, respectively.

TABLE III
SUMMARY OF IMPROVEMENTS ACHIEVED BY OUR PROPOSED METHOD AND
COMPARISON WITH STATE-OF-THE-ART

Method	Overall Accuracy (%)	MEM (KB)	OPS (K)
TBioCAS 2020 [16]	99.2	57.25	-
TBioCAS 2023 [17]	99.16	73	-
TCAS-I 2023 [27]	90.5	1.56	784.375
Our 2-D SNN	98.28	24.33	53.53
Our 1-D SNN	97.5 (-0.78%)	12.26 (2.02x)	26.52 (1.98x)

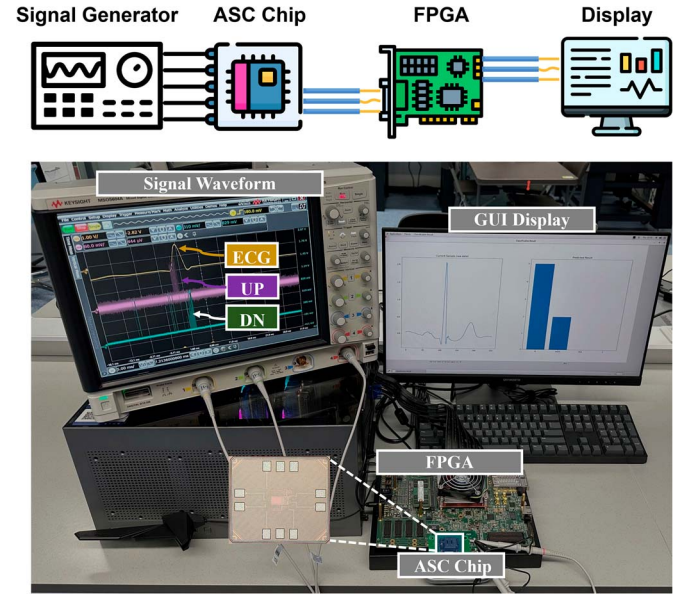


Fig. 17. FPGA-based testbench system of BioPI with a 40nm ASC chip.

The detailed comparison of this work with state-of-the-art is summarized in Table III. Compared to existing methods, our 2-D SNN demonstrates superior performance in terms of accuracy and computing efficiency with 24.33KB MEM and 53.53K OPS. Furthermore, our 1-D SNN achieves a further speed-up of 2.02x and 1.98x in MEM and OPS saving, respectively, with only a 0.78% decrease in accuracy compared to the 2-D SNN.

F. System Testbench

Fig. 17 illustrates the FPGA-based testbench system for BioPI. The raw ECG signal is generated from the signal generator and transmitted to the ASC chip. The UP and DN spikes are then extracted for testing, and the merged spike sequence is fed into the FPGA platform (Xilinx Virtex XCVU13P). The SPU in BioPI is implemented in Verilog and deployed on the FPGA. The classification results are displayed on a connected monitor using a dedicated GUI. Table IV summarizes the FPGA resource utilization of the SPU. As shown, the SPU occupies only 40642 look-up tables (LUT) and 67578 flip-flops (FF), highlighting the hardware efficiency of BioPI.

TABLE IV
FPGA RESOURCE UTILIZATION ON XILINX XCVU13P

Resource	LUT	FF	BRAM	URAM	DSP
Utilization	40642	67578	0	0	0

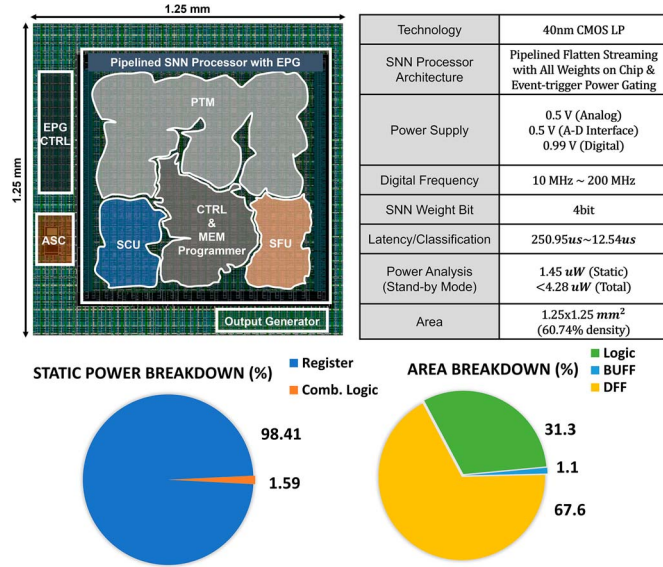


Fig. 18. The BioPI SoC layout in 40nm: physical view and specifications.

G. Silicon Layout Evaluation

On top of end-to-end functionality verification platform including a fabricated 40nm ASC chip and a FPAG implementation of the SNN processor, we implement the complete design of BioPI in 40nm process and conduct post-layout simulations to evaluate the overall energy efficiency of the entire system on silicon. The complete system-on-chip (SoC) with the pipelined flatten streaming architecture is integrated following the digital-top mixed-signal design flow, utilizing the Cadence Innovus Toolbox, with power and area reported. The final layout of the SoC is illustrated in Fig. 18. Post-layout simulations are performed to verify the functionalities of the BioPI SoC. Detailed specifications, including static power breakdown in stand-by mode and area breakdown of the SNN processor, are extracted and reported. The layout area measures 1.5625mm^2 with a density of 60.74%. The static power breakdown in stand-by mode is as follows: registers account for 98.41%, while combinational logics account for 1.59%. In terms of area breakdown, sequential cells (DFF), combinational cells (Logic), and buffer/inverter (BUFF) account for 67.6%, 31.3%, and 1.1%, respectively.

Fig. 19 presents the total power and energy consumption of the BioPI SoC in operation mode and the total execution time per classification with 10 steps for SNN inference. The power consumption ranges from 3.1mW to 61.4mW as the operating frequency varying from 10MHz to 200MHz. Concurrently, the total execution time decreases from 250.95us (operating at 10MHz) to 12.54us (operating at 200MHz). When operating at 200MHz, the inference energy consumption is 0.767uJ per classification. The energy savings brought by adopting the proposed

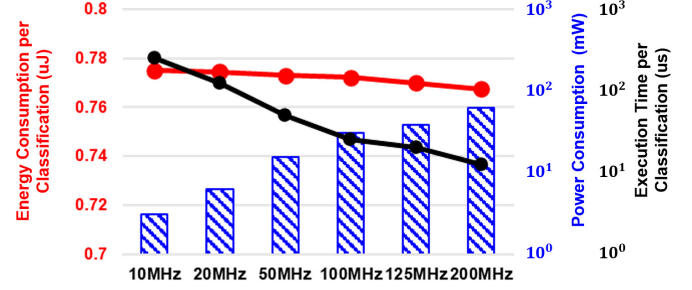


Fig. 19. Total power of BioPI SoC in operation mode and energy consumption with 10 time steps ($T=10$) for SNN inference.

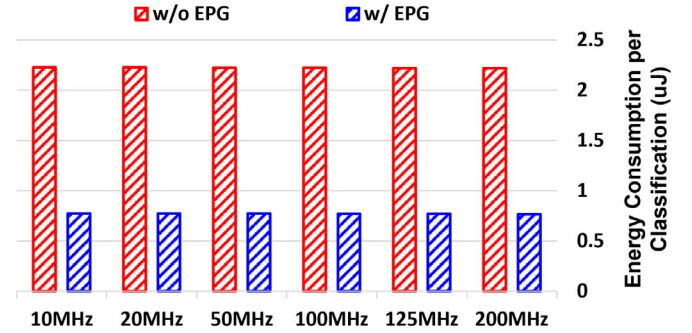


Fig. 20. Energy savings brought by adopting the proposed EPG scheme in BioPI SoC.

EPG scheme are illustrated in Fig. 20, showing more than x2.87 savings in energy consumption per classification.

H. Comparison With State-of-the-Art

Table V summarizes the comparison of BioPI with other state-of-the-art works in various metrics. BioPI exhibits advancement by adopting an in-sensor spike-based architecture, eliminating the need for an extra signal feature extractor. Designs in [16], [17], [23], and [26], are all based on homogeneous architecture, where the processor utilizes large parallel cores shared among layers. Although this common design is flexible by sequentially carrying out neural operations layer by layer, it is challenged by the varying size of layers in terms of latency and is rarely optimized for mapping an application-specific NN. Furthermore, these designs still require extensive off-chip memory access and a host CPU to control the scheduling for different layers, which sets limits for the bursting of processing efficiency. In [27], an asynchronous flatten streaming SNN is reported. Although low latency is achieved, the classification performance is unsatisfactory due to leakage and mismatch issues, as computation is performed in the analog domain. In this work, BioPI achieves an optimal point between the trade-off of classification performance, latency, and processing efficiency through a joint design optimization of sensor-algorithm-processor. More specifically, our design achieves 97.5% overall accuracy on 4-class ECG classification with only 4-bit weight resolution and consumes only 0.767uJ energy and 12.54us latency per classification when operating at 200MHz.

TABLE V
COMPARISON WITH STATE-OF-THE-ART WORKS ON MIT-BIH DATASET

Reference	TBioCAS 2019 [23]	TBioCAS 2020 [16]	TBioCAS 2022 [26]	TBioCAS 2023 [17]	TCAS-I 2023 [27]	This Work	
Sensor-Interface Integration	No	No	No	No	No	Yes	
Extra Feature Extractor	Required	Required	Required	Required	Required	Not Required	
Spike-Only Dataflow	No	No	No	No	Yes	Yes	
Off-Chip Host CPU	Require	Require	Require	Require	Not Require	Not Require	
Hardware Architecture	Homogenous (Parallel Cores)	Homogenous (Parallel Cores)	Homogenous (Parallel Cores)	Homogenous (Parallel Cores)	Flatten Streaming (Cascaded Cores)	Flatten Streaming (Pipelined Cores)	
Network Type	Spike-MLP	MLP	Spike-RNN	CNN	Spike-MLP	Spike-CNN	
Network Depth	5 Layer	3 Layer	4 Layer	4 Layer	4 Layer	3 Layer	
Weight Bit Width	-	16	6	16	4	4	
Accuracy	97.9%@2-class	99.2%@4-class	98.22%@5-class	99.16%@5-class	90.5%@2-class	97.5%@4-class	
Precision	97.3%@VEB	98.0%@VEB, 94.3%@SVEB	-	98.2%@VEB, 87.3%@SVEB	89.2%@N	99.4%@N, 95.9%@VEB, 96.8%@SVEB, 99.7%@F	
Specificity	99.8%@VEB	-	-	99.8%@VEB, 99.9%@SVEB	92%@N	99.2%@N, 98.6%@VEB, 99%@SVEB, 99.9%@F	
Hardware Evaluation	Theoretical Estimation	Post-Layout Simulation	Post-Layout Simulation	Silicon Processor	Silicon Processor	Silicon ASIC + FPGA Processor	SoC Post-Layout Simulation
Technology	-	180nm	40nm	65nm	180nm	40nm ASIC + 16nm FPGA	40nm SoC
Frequency	-	10KHz~25MHz	65KHz~100MHz	2.5MHz	Asynchronous	10MHz~200MHz	
Latency/Classification	-	629.8ms~251.92us	775.6ms~504.14us	41ms	40us	250.95us~12.54us	
Efficiency (Energy/Classification)	1.78uJ (Theoretical Estimation)	6.77uJ~3.21uJ (Processor)	20.51uJ~0.75uJ (Processor)	4.36uJ (Processor)	-	For Real-Time Functionality and Accuracy Testing	0.775uJ~0.767uJ (SoC)

V. DISCUSSION

This section aims to discuss the advantages and limitations of our proposed DM-BPTT algorithm and BioPI architecture. Although various algorithms adapted from BPTT have been reported [18], our proposed DM-BPTT algorithm shows superiority in efficiency and scalability by taking both the sensor and processor into consideration. Motivated by biological neural systems, DM-BPTT enhances inference efficiency by converting 2-D operations into 1-D operations and achieves state-of-the-art classification performance. Although more training effort is required in DM-BPTT compared to typical BPTT due to the introduction of a unique dual-neuron structure, the network size is always small for edge biomedical applications, where a modern CPU is sufficient to train the network within several hours. As DM-BPTT follows a similar computing graph as the original BP, the proposed dual-neuron structure could be scaled and deployed to multiple networks for various scenarios, while more training effort would be required as the application complexity increases.

In terms of hardware architecture, BioPI differs from popular homogeneous processors with parallel cores, as it utilizes a flatten streaming processor with pipelined cascaded cores. Homogeneous processors with reconfigurable processing elements currently dominate the design of NN accelerators [41], where extensive parallel arithmetic cells are used to form one computing core, and both weights and activations are buffered and sent into the core from on-chip memory. The computing core is time-shared as different NN layers are mapped separately by reusing the same hardware. However, the huge burden of

off-chip memory access for layer remapping and the requirement of a host CPU for complex scheduling greatly limit its performance in tackling application-specific tasks. Alternatively, BioPI divides different layers into different hardware modules and pipelines between modules to maximize the processing speed and hardware utilization ratio, which greatly accelerates the overall processing efficiency. Our proposed BioPI architecture provides a parameterized implementation of different kinds of layers, which can be easily scaled up and customized for various network structures. Since all the network parameters are stored on-chip in BioPI, neither off-chip memory access or host CPU is required. However, if the intended network is deeper with a much larger number of parameters, the on-chip hardware costs will also greatly increase. Thus, the proposed hardware architecture in BioPI is more suitable for the end-to-end implementation of tiny networks targeting edge applications such as healthcare monitoring.

VI. CONCLUSION

This paper introduces BioPI, a sensor-direct neuromorphic system, from a joint optimization perspective of sensor, algorithm, and processor. The entire system is implemented in 40nm CMOS process and verified on arrhythmia detection tasks using the MIT-BIH ECG dataset. Inspired by biological neural systems, BioPI employs fully-spike dataflow with a hierarchical pipelined architecture, combining a mixed-signal analog-to-spike converter and a compact flatten streaming SNN processor. A unique DM-SNN algorithm is also presented, which greatly reduces on-chip memory size and operation numbers

compared to existing approaches, while maintaining state-of-the-art accuracy. In summary, our BioPI offers several advantages, including accurate and efficient end-to-end on-chip inference for in-sensor integration. The proposed DM-SNN algorithm and pipelined flatten streaming hardware architecture are also scalable to other edge applications. Our experimental results validate the effectiveness of the proposed system and highlight its potential for future research in wearable healthcare applications.

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